THE NON-MINIMALITY DEGREE OF THE TYPE OF A POINT OUTSIDE THE PREDICATE IS R MINUS ONE

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Editorial note. Assembly history: r001 failed with 'LaTeX Error: Option clash for package graphics' because the author's own preamble loads \usepackage[dvips]{graphics,epsfig} while the wrapper class already loads graphicx without options. r002 splits the author preamble block into two spans and omits that single package line; no mathematical macro, statement, span or stub changed. Same family as the earlier \smaller

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clash: a publisher package line that must not be re-injected on top of the wrapper baseline.

Allowed standard prerequisites (editorial).

- The theory T_{r,k,mu} and its predicate P, with the relation R on r-tuples of points outside P. Set up in the construction preceding the statement; used inside the counted proof at lines 2354-2355. Supplies the tuple whose existence witnesses the non-forking chain and the relation controlling when a chain has a forking extension.
- The action of the automorphism group of the model is i-transitive on the complement of P. Used inside the counted proof at lines 2360-2362. Makes each type p_i, for i < r, the unique non-forking extension of p, which is what keeps the first r-1 steps non-forking.

Theorem 0.1. For $T_{r,k,\mu}$ the complete type over $\emptyset$, p, given by $\neg P(x)$ has nmdeg$(p) = r - 1$.

Proof. Choose distinct $a_1, a_2, a_3, \ldots a_r$ realizing $p(v)$ and such that $R(a_1, a_2, a_3, \ldots a_r)$. Let $p_i = \text{tp}(a_i/A_{<i})$. Each p_i for $i < r$ does not fork over the empty set because Aut(M) acts i-transitively on $\neg P$ guaranteeing each p_i is the unique non-forking extension of p to its domain $A_{<i}$. The type p_r forks over $\emptyset$ because there are infinitely many distinct choices of a^j_{r-1} giving different lines. Any pair of the lines $\ell_i = \langle a_1, a_2 \ldots a_{r-1}, a^j_r \rangle$ are infinite and intersect only in $a_1, a_2 \ldots a_{r-1}$ since any further point of intersection would imply the lines are identical. □